IMPACT OF BARBECUED MEAT CONSUMED IN PREGNANCY ON BIRTH OUTCOMES ACCOUNTING FOR PERSONAL PRENATAL EXPOSURE TO AIRBORNE POLYCYCLIC AROMATIC HYDROCARBONS. BIRTH COHORT STUDY IN POLAND

We previously reported an association between prenatal exposure to airborne PAH and lower birth weight, birth length and head circumference. The main goal of the present analysis was to assess the possible impact of co-exposure to PAH-containing of barbecued meat consumed during pregnancy on birth outcomes. The birth cohort consisted of 432 pregnant women who gave birth at term (>36 weeks of gestation). Only non-smoking women with singleton pregnancies, 18-35 years of age, and who were free from chronic diseases such as diabetes and hypertension were included in the study. Detailed information on diet over pregnancy was collected through interviews and the measurement of exposure to airborne PAHs was carried out by personal air monitoring during the second trimester of pregnancy. The effect of barbecued meat consumption on birth outcomes (birthweight, length and head circumference at birth) was adjusted in multiple linear regression models for potential confounding factors such as prenatal exposure to airborne PAHs, child's sex, gestational age, parity, size of mother (maternal prepregnancy weight, weight gain in pregnancy) and prenatal environmental tobacco smoke (ETS). The multivariable regression model showed a significant deficit in birthweight associated with barbecued meat consumption in pregnancy (coeff = −106.0 g; 95%CI: −293.3, −35.8); The effect of exposure to airborne PAHs was about the same magnitude order (coeff. = −164.6 g; 95%CI: −172.3, − 34.7). Combined effect of both sources of exposure amounted to birth weight deficit of 214.3 g (95%CI: −419.0, − 9.6). Regression models performed for birth length and head circumference showed similar trends but the estimated effects were of borderline significance level. As the intake of barbecued meat did not affect the duration of pregnancy, the reduced birthweight could not have been mediated by shortened gestation period. In conclusion, the study results provided epidemiologic evidence that prenatal PAH exposure from diet including grilled meat might be hazardous for fetal development.